
Homework 5

Grant Talbert

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Boston University

Problem 1

Let r be a root of $f(x)$ in $\mathbb{C}$, viewed as an extension field of $\mathbb{Q}$, and note that $r \neq 0$:

$$0^4 + 0^3 - 0^2 - 0 + 1 = 1 \neq 0.$$

Therefore, since $\mathbb{C}$ is a field, $-1/r$ exists. We have

$$\begin{aligned} \left(-\frac{1}{r}\right)^4 + \left(-\frac{1}{r}\right)^3 - \left(-\frac{1}{r}\right)^2 - \left(-\frac{1}{r}\right) + 1 &= \frac{1}{r^4} - \frac{1}{r^3} - \frac{1}{r^2} + \frac{1}{r} + 1 \\ &= \frac{1}{r^4} - \frac{r}{r^4} - \frac{r^2}{r^4} + \frac{r^3}{r^4} + \frac{r^4}{r^4} \\ &= \frac{1 - r - r^2 + r^3 + r^4}{r^4} \\ &= \frac{r^4 + r^3 - r^2 - r + 1}{r^4} \\ &= \frac{f(r)}{r^4} \\ &= \frac{0}{r^4} \\ &= 0. \end{aligned}$$

Therefore, $-1/r$ is a root of f .

Problem 2

By problem 1, roots come in pairs of the form $(r, -1/r)$. By a theorem I found on Google, the product of all roots of f is the constant term of f . By the method of proof by the professor said so, r exists, and thus f does not have 0 roots. Suppose f has 1 root. Then $r = -1/r$, and thus $r^4 = 1$. Since the only fourth roots of 1 are $\{\pm 1, \pm i\}$, and since none of these are roots of f :

$$1 + 1 - 1 - 1 + 1 = 1, \quad 1 - 1 - 1 + 1 + 1 = 1,$$

$$1 - i + 1 - i + 1 = 3 - 2i, \quad 1 + i + 1 + i + 1 = 3 + 2i,$$

we get a contradiction. Therefore, f has more than 1 root. This also shows that roots need to come in pairs with *distinct elements*, since $r \neq -1/r$ for r and root, and thus the roots must be an even number. Since f has degree 4, it can have at most 4 roots. Therefore, it has either 2 or 4 roots.